Business Horizons (2024) 67, 511—523

Available online at www.sciencedirect.com

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Managing a ChatGPT-empowered workforce: Understanding its affordances and side effects

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KEYWORDS
Generative AI;
ChatGPT;
Large language models;
Affordances;
Knowledge work

Abstract Generative AI, particularly ChatGPT, creates new managerial concerns. This article addresses a crucial managerial challenge: While employees are including the tool as their private knowledge assistant in many aspects of their daily work, it is difficult for managers to see and understand the impact within their organization. There is more and more literature on what ChatGPT means for business, but we cannot say much as long as we are uninformed about its actual use. So, getting an insider's perspective is needed to think about the consequences for organizations. The article is grounded in a qualitative study of employees' experiences of interacting with ChatGPT among 50 early ChatGPT adopters. First, we show how employee-ChatGPT relations develop from private, experimental use into an integral part of knowledge work. Second, we identify six affordances: searching for information, brainstorming ideas, structuring content, writing a first draft, embellishing text, and proofing work. Further, we highlight three looming side effects of ChatGPT use that threaten knowledge ties, the quality of knowledge in organizations, and how work roles are configured. Accordingly, this article offers managers guidance in managing a ChatGPT-empowered workforce in a way that aims to mitigate these side effects while harnessing opportunities.
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https://doi.org/10.1016/j.bushor.2024.04.009
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1. Why is employee use of ChatGPT a managerial concern?

Generative AI (GenAI) tools and specifically ChatGPT are seeing rapid global adoption (Hu, 2023). Adding a chat interface made using large language models intuitive even for those with novice levels of digital literacy (Kietzmann & Park, 2024). GenAI has landed in industries such as innovation and R&D development (Sundberg & Holmström, 2024), customer service (Ferraro et al., 2024), marketing (Cui et al., 2024; Osadchaya et al., 2024), and higher education (Hashmi & Bal, 2024). However, managers are still in the dark about how their employees turn to ChatGPT (Budhwar et al., 2023). A recent estimate is that 2 in 3 employees use ChatGPT without their manager knowing (Christian, 2023). Understanding how employees use ChatGPT is important yet difficult because GenAI is different from traditional AI (see Table 1). Whereas traditional AI can recognize patterns and optimize classification and decision-making (Leyer & Schneider, 2021; Vincent, 2021), ChatGPT generates new output (Zhang et al., 2023) which may be indistinguishable from human work, though it can 'hallucinate' (Bang et al., 2023; Hannigan et al., 2024). Furthermore, OpenAI's democratizing open access release resulted in a bottom-up introduction process of ChatGPT, in contrast to traditional AI's top-down introduction (e.g., Waardenburg et al., 2021; Strich et al., 2021).

To understand how employees use ChatGPT, we conducted a qualitative study. We learned that it increased worker productivity, but we also identified side effects that are important for managers (Zammuto et al., 2007). First, we show how ChatGPT jumps from private, experimental use into an integral part of knowledge work. We identify in our interviews six affordances of ChatGPT in knowledge work. The affordance lens directs us to what ChatGPT allows employees to do in practice (Leonardi, 2011), as opposed to the designed features or capabilities of a technology

Table 1. Generative AI vs Traditional AI: Similarities and differences

	Traditional AI	Generative AI
Keywords	"automation" "prediction" "accuracy" "objectivity"	"generation" "productivity" "efficiency" "synthesis"
Bias	Problematic but often statistically detectable	Subtle and often requires subjective interpretation, and danger of hallucination
Regulatory reaction	Slow and methodical, consultative (guidelines)	Radical and rapid (e.g., Italy banning)
Development speed	Fairly slow	Extremely rapid (e.g., new applications and plugins launching daily)
Implementation process	Top-down implementation. It is too complicated and expensive to initiate bottom up, managerial initiative and investment required	Bottom-up adoption (comes from personal sphere, third party developments are so fast and consumer-driven, difficult for management to get an overview)
Interface	Workers often did not interface directly with tool, more likely to be provided outputs while other parties had access to other layers	Text based interaction, everyone has similar access to tool/format of interface is similar/same for all (prompting)
Affordances	Narrow, predefined	Multipurpose, open-ended
Transparency	Fairly easy to detect because it is bad at imitating human-like content	Difficult to detect because it is exceptional at imitating human-like content (natural speech, images)
Relationship to probabilistic logic	Overtly probabilistic (e.g., often presenting a probability as output)	Covertly probabilistic (e.g., seems to create responses) but output is generated through probabilities (e.g., of next word)

(Faraj & Azad, 2012). Affordances describe how users actually use a technology (Gibson, 1977). This is particularly relevant in the early stages of ChatGPT use, because the technology is radically open-ended and people are still figuring out what it can do.

Second, we show various side effects of ChatGPT use and infer their implications for management. *Side effects* describe the broader, long-term, and often unexpected, consequences of technology use (Zammuto et al., 2007). For example, we found that employees go to ChatGPT instead of to colleagues for advice. This is problematic because within-firm knowledge sharing and integration is a core ingredient for organizational learning and competitive advantage (Azeem et al., 2021) and arguably the fundamental purpose of organizations (Grant, 1996; Spender, 1996). If coworkers do not go to each other, then why even have an organization to bring them together? Drawing upon and extrapolating from our study insights, we show managers how ChatGPT can benefit the knowledge production and application in their organization (Jarrahi et al., 2023; Ramaul et al., 2024) while also highlighting the (unintended) side effects for their organization. We advise managers on how to mitigate the problematic aspects of these side effects to ensure successful knowledge management in times of ChatGPT (Feng et al., 2024).

2. The value of the affordance lens

New technologies are seldom used in a way that managers and/or designers intended. What actions a technology enables or constrains people to do, cannot be captured by simply looking at the designed properties or features in isolation. Affordance theory is insightful because it brings to the fore how people use a technology in practice, highlighting that such uses can be unexpected and change over time (Schultze & Boland, 2000). Thus, it is the interaction of humans and technologies doing things that shape what actions actually emerge; in other words, it is the relation between human and technology that affords the actions (Leonardi, 2023). For example, a nonobvious affordance of social media is that its use acts as a leaky pipe for knowledge, because conversations between two people can be visible to the organization (Leonardi et al., 2013). Another example is the printer room, a nonobvious affordance of which is stimulating informal interactions between colleagues. If the printer is located where people experience privacy, it gives workers a justification to be there and not look lazy while chatting (Fayard & Weeks, 2007). Identifying this nonobvious affordance has helped designers of office spaces use this affordance to stimulate collaborative work environments (Colenberg et al., 2023).

As a large language model, ChatGPT is malleable, multipurpose, and learns autonomously (Schmitz et al., 2016). Therefore, we need to understand its affordances—what workers actually do with ChatGPT—to capture its impact on work and organizations (Leonardi, 2011). In this article, we then go a step further by asking: Given these affordances, what side effects can managers anticipate and possibly, where needed, mitigate?

3. Interview study with early ChatGPT adopters

We conducted a qualitative interview study with 50 early adopters of ChatGPT (Dedehayir et al., 2017; Rogers, 2003). Our data collection took place from January until July 2023. We used purposeful sampling to recruit people who had interacted with ChatGPT. We found participants by searching on LinkedIn via the hashtag #ChatGPT to see people who had posted about ChatGPT, whether in positive, negative, or neutral terms. We sent interview requests to them, where we highlighted that we welcomed all levels of usage in our participants, from irregular to routine users. Our data is unique in terms of exploratory insights because all 50 participants were from 50 different organizations, offering a broad view on what is happening across organizations, yet all engaged in knowledge work, such as project management, research, marketing, copywriting, UX design, IT/programming, investing, recruitment, customer service, and consulting. Further, because we were not part of the participants' organizations, they trusted us with openly sharing their use experiences with ChatGPT. Interviews followed an explorative protocol with broad questions on user experiences related to current and past use of ChatGPT as well as their perspectives on ChatGPT. Interviews lasted between 32 and 102 minutes, with an average of 70 minutes. We recorded all interviews with the participants' consent.

We analyzed the data iteratively, adopting an abductive research methodology where we went from our data to relevant theory and back to our data to make sense of it more thoroughly (Charmaz, 2014). We had peer debriefing as a form of investigator triangulation (Archibald, 2016), whereby the researcher who conducted the interviews proposed codes and the co-authors proposed sharper or more nuanced versions. For

example, we first paid attention to the way users start using ChatGPT: what motivated them to interact with it and how they experienced these interactions. Second, we grouped how they came to use ChatGPT in their work practices, which we then understood and labeled as the current affordances of ChatGPT for knowledge work. Although these affordances are still somewhat malleable (Schmitz et al., 2016), we focused on affordances about which there was a degree of closure; that is, a stage of an innovation where groups of users settle on what constitutes its appropriate use (Bijker et al., 2002).

4. An insider's perspective: ChatGPT's trajectory from private to work contexts

4.1. Building the relation: From playful tinkering to purposeful use

We found that early adopters were driven to use ChatGPT either out of curiosity or anxiety. While some were intrigued by the media hype around GenAI, others mentioned anxiety that soon their organization may not need their human labor anymore. Trying out ChatGPT was sometimes seen as a way of getting ahead of this threat. For example, participants expressed the need to now get on board with these developments, to evolve with them. Becoming an expert user was seen as a strategy to avoid becoming replaceable.

Whether out of curiosity, anxiety, or both, we found that participants start using ChatGPT through tinkering. *Tinkering* refers to the activity of interacting with a technology in an iterative, trial and error manner. It involves hands-on exploration and adaptation, to figure out new solutions to problems that arise in the flow of use (Ciborra, 1992). Because use was inspired by an emotional response to the technology rather than a directive from management, participants tinkered regarding topics that popped up in their private lives. For example, they described asking for advice on what to talk about with children, how to handle a teenage son, and where to travel. They tinkered in the context of dumb stuff (e.g., a marketing professional reported asking ChatGPT to write poems in the style of a famous poet, about the user's hair).

At face value, such private-life tinkering may seem irrelevant to managers. However, participants described that this tinkering helped them understand ChatGPT. The same marketing professional noted: "And then you begin to see, what can I get out of it? What does that thing do? What can one even do with it?" Participants articulated that after having played with ChatGPT for some time, they used it in a much more targeted way for outsourcing their tasks because they had identified what ChatGPT could do for them.

Tinkering in a playful, experimental fashion taught users what the technology could do, but also what its limitations are and what it should not be asked to do. ChatGPT often hallucinates; that is, it confidently presents as truth information that is factually incorrect. Users learned to detect such inconsistencies and became mindful of using ChatGPT for fact finding. Participants explained discovering hallucinations when checking generated information (e.g., seeing that academic sources provided by ChatGPT did not exist).

In sum, the understanding of ChatGPT's strengths and limitations was not gained through a user manual or through training, and at the time of research, also not yet through the media but through first-hand tinkering in both private and professional spheres. After understanding why early adopters started using ChatGPT (curiosity and/or anxiety) and how tinkering helped them to see what ChatGPT offered them, we saw that participants drew on these experiences to identify key affordances of ChatGPT for their daily work practices.

4.2. Reaching intertwinement: ChatGPT as a number one knowledge assistant

We heard that ChatGPT increasingly became a part of our participants' daily work. We describe this stage, where ChatGPT becomes part of normal flow of work, as *intertwinement*. The idea of intertwinement is inspired by literature that looks at how people relate to technologies that they are familiar with and reliant upon. Reading glasses and a blind person's cane are classic examples where it is not clear where the human ends and the technology begins; they operate as one closely linked, intertwined entity (Merleau-Ponty et al., 2013). Likewise, over time ChatGPT became our participants' go-to place for assistance. Some noted they use ChatGPT every day and always have a browser window open, ready to chat. It became an indispensable routine part of their work, and participants referred to it as "my personal assistant" or "it's a part of my workflow." In this stage, the tool becomes such an integral part of the work that it is no longer clear to the

user what aspects of their work are which are human and which are AI generated.

We further noticed that when the worker-technology relation reaches this intertwinement stage, participants reported working with ChatGPT similar to how they would work with a colleague. For basic tasks, they feed ChatGPT with the kind of instructions and context that would often be given to a junior colleague. More complex, higher-level questions were also posed to ChatGPT. For example, instead of turning to a business partner for help with a task she is known to be proficient in, ChatGPT was seen as a more efficient place to turn.

Participants in this stage of intertwinement made comments that are reminiscent of dependence or even addiction. Turning to ChatGPT, some participants reported, became a new normal for them. In this way, ChatGPT is a double-edged sword for employees because they became so reliant on it that being without it becomes a problem. A consultant stated:

> I feel more professional and engaged. It helps me with my work efficiency and quality a lot. But also, I feel a bit more stupid. I feel like very much depending on this technology. Like right now, if you ask me to write something, I start to feel scared. I don't really trust my behavior. I have to rely on this technology.

4.3. ChatGPT's affordances for knowledge work

The worker-technology relation gave rise to affordances. We paid attention to participants' descriptions of how they used ChatGPT in their work and bundled these descriptions into *affordances* of ChatGPT for knowledge work, shown in Table 2. Each affordance is concretely illustrated with example prompts for ChatGPT to assist with a marketing campaign.

Some participants primarily made use of a few affordances while other proficient users described jumping between the affordances very fluidly (e.g., searching for information, structuring an outline, and generating a first draft within a single chat thread).

4.3.1. Searching for information

Participants often compare ChatGPT with the Google search engine, saying they used Google before but now go to ChatGPT if they need to find information. Several participants appreciated the quick and easy access to tailored information offered in a digestible format. One consultant highlighted: "ChatGPT brings a tremendous knowledge base to the table." We heard that ChatGPT offers a more precise answer than Google and it can also be asked to expand on details that

Table 2. ChatGPT affordances: What participants use ChatGPT for

Affordance	Description	Illustrative prompts: Creating a marketing campaign proposal
Searching for information (37 participants)	Finding information; Seeking advice; Filling blanks	Please conduct research on client xyz's industry. Identify key trends, opportunities, and challenges that impact the industry's growth and potential.
Structuring content (35 participants)	Asking to summarize content; Asking for an outline	Please summarize the following marketing report into one paragraph, followed by three key takeaways.
Brainstorming ideas (36 participants)	Asking for initial ideas and options that are then narrowed down and expanded upon	Give me five suggestions for a marketing campaign slogan that is timely and memorable.
Writing a first draft (34 participants)	Creating an initial, basic version of a draft that can be edited	Write me a project proposal for a marketing campaign, and include the following information…
Embellishing text (16 participants)	Adding flourishes to existing text	Add descriptive adjectives to the following copy, in the style of a 1950s advertising campaign.
Proofing work (25 participants)	Feeding work in to check if spelling or grammar is correct; Checking for tone; checking and correcting for bugs in computer code	Please check whether the campaign proposal is in professional English, with appropriate use of slang for a creative industry context

the user found relevant (this is what UX designers refer to as *progressive disclosure*). We heard that it is better to search within one's area of expertise, because it is then easier to discern whether the results are likely to be accurate, or false hallucinated information.

Participants reported turning to ChatGPT for work-related information they would otherwise seek from their manager or colleagues. As a copywriter said:

> It feels a bit reassuring to be honest, because I don't have to run up to my manager every two seconds to tell them I don't know how to do it. At first of course I try to research. And I try to like, for example, I don't know how to write an ad copy. So I can just go to ChatGPT. Like, how would you write an ad copy? And it would give me examples.

4.3.2. Structuring content

ChatGPT is also used to structure content (e.g., by requesting a summary or bullet point outline of text that was fed to the system). This affordance of structuring content makes it possible for users to grasp (apparent) key ideas from text that was either deemed to be too long and/or too complicated to digest. Thereby, participants reported now being able to comprehend information that they otherwise would not because the information lies outside of their expertise, such as an investor understanding a scientific article or a consultant checking clauses of legal documents. Participants felt that this is a fast and efficient process that makes them less dependent on others.

Participants also describe using ChatGPT to structure content they are working with into another format or genre. A consultant explained that she first writes down all her thoughts and ChatGPT helps to structure collected content within a specific thinking framework that she has read about but does not know how to apply herself. She explained:

> So without reading a book about it, without knowing it, I can apply these different well-known methodologies [i.e. frameworks for consulting] in the way I structure my thoughts. Typically, my business partner comes and puts the structure in based on his knowledge of frameworks, but I actually don't need him anymore.

Again, being more efficient in knowledge management processes such as knowledge retrieval, knowledge transfer, and knowledge application also means being less reliant on others' help.

4.3.3. Brainstorming ideas

Participants reported that they use ChatGPT for exchanging ideas back and forth, just like brainstorming with a colleague. A UX designer said:

> It can be pretty good at coming up with ideas.... It's the same thing as getting a group of people in a room and saying, let's, you know, whiteboard something…. And brainstorming again, you know, I could spend a day with my team coming up with ideas.

ChatGPT is accessible 24/7 and always happy to help if employees are not around or have no inspiration. Apart from helping them to get going this participant notes that ChatGPT can help refine their ideas. We heard that participants could come up with more ideas, more quickly, and could then bring these options to the team.

4.3.4. Writing a first draft

Participants said they have trouble getting started on written work, and ChatGPT helps them overcome the blank page. Participants said it saves them a lot of time because their threshold to get started is much lower. Furthermore, using ChatGPT for a first draft also reportedly improves work output. For example, our participants highlighted that they work more efficiently because they can focus on checking facts and tweaking the text instead of needing to create all of it from scratch. Essentially, their work shifted from writing to editing the output that ChatGPT provided to make it their own text. The time spent on writing a first draft was sometimes halved. In sum, using ChatGPT for a first draft speeds up the writing process and allows users to focus on refinement. Yet, it also shows the side effect that certain skills become more important (e.g., editing rather than writing).

4.3.5. Embellishing text

Embellishment describes using ChatGPT for the final touch on content (e.g., by adding descriptive details to creative writing or formalities to a letter). For instance, participants reported making a piece of text for social media with ChatGPT a little bit more lighthearted or adding in a joke. We also found that employees use embellishment to add niceties to an email to improve the tone of what was otherwise quite a harsh message. For example, a consultant said: "Tell this idiot colleague of mine how to do this, this, and this, and a very friendly email comes out." By adding the intended tone and style to a message, ChatGPT's embellishment can help employees make a text more tailored and appealing for the receiver.

Participants also use ChatGPT to embellish their prompts for other GenAI tools such as Midjourney or DALL-E that generate pictures based on text. To generate pictures that look real, camera details need to be put into the prompt. Hence, ChatGPT helps to add details to the prompt, to generate a more sophisticated picture.

4.3.6. Proofing work

We heard repeatedly that ChatGPT is used as a proof reader to check participants' work for any mistakes or potential problems, to give feedback, and to fix issues. For example, proofreading is done to check the accuracy of language—mainly grammar and tone—or whether they have missed important points, before sending out their emails or written documents. One UX designer said:

> Honestly, the main reason that I'm using ChatGPT is because I have a little fear about my language, especially with the writing. And, maybe I use my language in a way it could maybe hurt someone else's feeling. Because I lack understanding of the culture…As a non-native English speaker ChatGPT is my go-to tool. That makes me more confident.

ChatGPT is also reportedly used for fixing issues in computer code. Participants reported asking ChatGPT to resolve error codes effectively. This approach allowed for code adjustments based on ChatGPT's suggestions. Without ChatGPT, some participants would have needed to invest much more time, or ask a colleague with more advanced coding skills to find and fix the bug. In this way, ChatGPT is used to check for errors and refine output in the way that a senior colleague might look over an employee's work.

5. Side effects of ChatGPT use and implications for knowledge work

In the above we showed how ChatGPT traveled from the private sphere to becoming a number one knowledge assistant. The resulting affordances add understanding of how previously identified productivity gains are achieved (e.g., Brynjolfsson et al., 2023; Noy & Zhang, 2023; Peng et al., 2023). However, in the following, we outline three side effects that we foresee. They require management's attention because they are usually both unexpected and long lasting[1], and hit at an organizational scale.

[1] See, for example, Zammuto et al. (2007).

5.1. Side effect 1: Broken knowledge ties

Across affordances, we found that employees turn to ChatGPT instead of turning to a colleague. For instance, before ChatGPT was introduced, proofing or embellishment were often done by asking another colleague to validate or to fine-tune content. Brainstorming was an important team process. Such shifts toward isolated work are relevant for managers; what is not happening is just as important as what is happening. As a side effect of ChatGPT use, employees do not go to a colleague for help with their work. We argue this is extremely consequential for organizations because it interrupts their routines for socialization and idea development.

The foregone social interactions in everyday practice (Feldman & Orlikowski, 2011; Hansen et al., 2005) are key to managing especially tacit and informal knowledge (Hadjimichael & Tsoukas, 2019). As stated by Tsoukas and Vladimirou (2001, p. 991): "The management...of organizational knowledge implies more the sensitive management of social relations and less the management of corporate digital information." Through day-to-day collective practices employees absorb practical wisdom about how to do their work and gather knowledge about who knows what (Brown & Duguid, 1991). In particular, it is knowledge sharing via knowledge ties between employees (Alerasoul et al., 2022) that makes knowledge an asset for organizations (Ipe, 2003), underpinning the organizational learning and innovation from which a firm gains a competitive advantage (Kamaşak & Bulutlar, 2010; Nonaka et al., 2006; Singh et al., 2021). Crucially, we contend that such ties are being disrupted by ChatGPT use.

So, our findings suggest that on the near horizon is a future where work becomes more individualized and isolated. With hybrid working a norm, the risk of worker isolation is already high. Adding ChatGPT into the mix, we see the potential for expert knowledge to become siloed, and for new hires to lack opportunities for development.

5.2. Side effect 2: Limited managerial oversight of the quality of knowledge

Knowledge is a key organizational resource (Nonaka et al., 2006), so it is crucial to control its quality (e.g., accuracy, applicability). This is why organizations use technology-supported knowledge management systems such as content management systems or knowledge repositories to document knowledge, draw on it, and evaluate it (Davenport & Prusak, 1998). However, we found

that employees use ChatGPT for searching for information, for writing a first draft and validating work. A side effect of this is a danger of introducing and reinforcing low quality knowledge into the organization. Moreover, we found that over time employees become *intertwined* with ChatGPT. In this stage, it is no longer clear (even to the user) what aspects of their work are human and which are AI generated. As a side effect, it becomes difficult to control the quality of knowledge that circulates within the organization and is delivered to clients. For instance, two US lawyers submitted a brief to the court that included made up cases by ChatGPT (Weiser, 2023). In sum, while employees and managers may be charmed by the direct effect of increased knowledge work efficiency, it becomes difficult for managers to oversee and control knowledge production and application.

5.3. Side effect 3: Reconfigured work roles

Work roles are reconfigured as a side effect of ChatGPT use. For instance, in writing a first draft, employees' role shifts to being an editor. For embellishing text, the roles shift to being a director (e.g., a film director). Across affordances, employees' roles shift to decomposing tasks, instructing, and delegating. Therefore, managers can no longer rely on their traditional understanding of what it means to be a high performing employee for a particular role. So, assessing employee performance becomes more difficult. This is exacerbated by ChatGPT's involvement making it difficult to infer an employee's contribution from the work output. Moreover, due to intertwinement, employee's themselves cannot even tell what aspect of their work is human- or AI-generated. This also has implications for developing employees. Reconfigured roles make it more difficult for managers to know what employees need to learn, or need help with. On top of this, if managers try to adjust their employee evaluations to this, it can lead to employees feeling unequally assessed and supported because not all employees have unlocked a variety of affordances.

Outsourcing tasks to, or co-creating tasks with, ChatGPT (Glaser et al., 2024) leads some work roles to have very few tasks left. One work role that is particularly at risk is novices performing low-status, mundane tasks; that is, scut work (Huising, 2015). Several interviewees commented that ChatGPT produced the kind of work they would expect from a junior colleague or intern. Why, then, would employees ask an intern to do this scut work? While a standard retort may be that interns can now focus on more high-value work, the availability of entry level tasks is important because they give novices a legitimate reason to get involved in a work process; that is, legitimate peripheral participation (Lave & Wenger, 1991). That participation means access to learning opportunities such as feedback, seeing how people work and interact, and what differs between a first draft and the final product. Doing scut work builds the competency, credibility, and experience necessary for a novice to develop into an expert. Thus, the work role reconfiguration reduces opportunities for development of novices.

In Table 3, we synthesize the three side effects and their implications for managers and knowledge work, and connect them to actionable recommendations for managers.

6. Recommendations for managers

Regarding the use we observed, we advise managers to encourage employees' tinkering because this unlocks affordances. ChatGPT's affordances can be a resource for knowledge workers to more efficiently perform their activities. This helps mitigate the high job demands that knowledge workers often face such as work intensity, performance, and time pressure. In the following, we also offer three recommendations for managers to help them spot and mitigate the organizational drawbacks of the three side effects we foresee.

6.1. Repair and retain knowledge ties: Designing social interactions

Regarding broken knowledge ties, we advise managers to create opportunities for collaborative interaction at work—those kinds of interactions that benefit knowledge ties. This is particularly important in an age of hybrid working, where employees may miss each other and therefore be more inclined to turn to ChatGPT for fast responses. First, incentivize interaction that is related to work, such as collaborative working practices to encourage opportunities for workers to learn from one another. We do not advocate for interaction for the sake of interaction (i.e., forced fun). Interaction should be carefully tied into how managers expect work to be done. Second, ensure that interaction is live. Asynchronous working allows moments to ask ChatGPT, but real-time interactions are spontaneous and require quick independent thinking and on-the-spot decision making. It is important to ensure that such live interactions are occurring regularly and are

Table 3. ChatGPT's affordances: Effects, implications, and recommendations

	Affordances with side effects	Implications for knowledge work	Recommendations for managers
Side effect 1	ChatGPT offers immediate assistance with tasks (e.g., proofing work, embellishing text, brainstorming ideas); employees turn to it instead of to each other/managers	Broken knowledge ties: Isolated workers and siloed knowledge	Repair and retain knowledge ties: Designing social interactions • Facilitate social interactions • Discuss work routines • Encourage live collaboration
Side effect 2	Reliance on ChatGPT for searching information, writing a first draft, and validating (for example) is risky; intertwinement with ChatGPT means employees cannot tell where their knowledge ends and AI starts	Lacking oversight of the quality of knowledge: Hallucinated content can reach clients, lack of validation of knowledge, social checks and balances are missing	Get involved in order to set the standards • Use ChatGPT so that the possibilities and limitations are understood • Resource key users to provide guidance: partner a star user with a skeptical user • Keep GenAI policies dynamic and responsive to changes
Side effect 3	Skills and tasks required to perform knowledge work change (e.g., first drafts can be generated quickly but need to be edited carefully)	Reconfigured work roles: Assessing and developing employee performance becomes a key challenge as people's roles shift in response to changing capabilities	Look to the horizon: What do ChatGPT's affordances mean for work in the long-term? • Realize how ChatGPT affordances can relieve work pressure • Hold open conversations about ChatGPT's role in knowledge work • Revisit talent development initiatives and emphasize the importance of live experiences for new hires

appropriately facilitated with resources such as working spaces. Otherwise, employees can get rusty in the important arts of conversation, negotiation, and presentation.

- *Guiding check-in questions for managers*: What reasons and meaningful incentives am I giving my employees to work and socialize together? What resources are provided to support live, in-person collaboration? What conversations do we need to have about working routines to ensure that people from different functions have the opportunity to regularly interact?

6.2. Get involved to set the standards

Given the difficulty of maintaining oversight of knowledge quality, banning ChatGPT may seem like a quick fix. However, GenAI is not going to go away and competitors that do not ban it may gain an advantage. Further, for ChatGPT, policing is difficult. With intertwinement, workers can not accurately report on the use and contribution of the tool (Riemer & Johnston, 2014). Moreover, policing may even be counterproductive. Demonizing ChatGPT is a surefire way to increase siloed knowledge. Employees will then use it as a

shameful secret, further damaging knowledge ties. On the other hand, fully unrestricted use of ChatGPT can be risky due to, for example, hallucinations, legal issues such as unknowingly using copyrighted text, leaking sensitive data, or overreliance (Fui-Hoon Nah et al., 2023).

Therefore, we advise managers to acknowledge what ChatGPT can do for the organization (Berthon et al., 2024). Managers should tinker with it themselves and do so openly, discussing what they learn about its limits and possibilities with both other managers and with their employees (Robertson et al., 2024). This will set the tone that it is okay to consult ChatGPT, and that this can be done as part of a social process—which includes the usual checks and balances of social interaction—rather than (only) as a private practice. Thus, make ChatGPT a part of the team, rather than trying to unwittingly manage a gang of secret AI assistants.

Furthermore, we recommend enlisting a *star user* who is passionate about the technology and at the forefront of using it in the organization. As a balance to this enthusiasm, we suggest partnering such a star user with a *critical counterpart*: someone in the organization who is informed yet skeptical or concerned about the use of ChatGPT at work. Resourcing these two ChatGPT mentors to speak with each other and host conversations and workshops is a way to share what is possible (unlocking new affordances for the workforce) while retaining a responsible and considered attitude. This duo could also bring urgent issues to management attention, such as ethical or legal concerns (Abdelhalim et al., 2024) or new technical developments (Brewer et al., 2024; Earley, 2023).

- *Guiding check-in questions for managers*: Do I acknowledge that ChatGPT is already an assistant to many of my employees? Am I tinkering with the tool myself to explore what is possible, also as it evolves technically? What tone do I set when I talk (or don't talk) about ChatGPT? Who can I empower to lead ChatGPT understanding in the organization?

6.3. Rethink ways of assessing and developing employees

In light of work role reconfiguration through ChatGPT, managers might aim to prevent reconfiguration to conserve efficient and standardized interrole routines. However, we advise managers to empower employees' responsible job crafting with ChatGPT, and to actively co-create reconfiguring work roles strategically. Given what ChatGPT can now do quickly and easily, what work objectives need to change? For example, translation of documents has been an important and time consuming component of many workflows. Yet, ChatGPT can perform an initial translation of text, tailored to particular audiences and language levels. What changes to work roles and objectives does such an affordance invite? Experts are still required to check and edit such automated content. This may prompt a reshuffling of who is needed to do what and when. By empowering and actively co-creating role reconfiguration, managers can help more evenly distribute ChatGPT use across the workforce (rather than some gaining an advantage).

Furthermore, regarding the risk that new talent is left underdeveloped, and no one is learning the unique way that things are done in the organization, we advise managers to not forget their junior staff on entry level positions and interns. It is important for organizations to change the learning and development programs to adapt to new needs and gaps that are emerging in socialization processes to secure a talent bench and succession. A concrete example for securing the talent bench is designing mentoring programs where the importance of tagging along to, for example, meetings and presentations is highlighted. In addition, to ensure succession, managers are advised to be cautious about downsizing their entry-level workforce in spite of the appearance of lower workload for them due to the efficiency gains from ChatGPT's affordances, focusing instead on forecasting in demand skill sets that need to be brought in and developed.

- *Guiding check-in questions for managers*: How can roles be adjusted over time to make the most of human expertise? How might we need to change our learning and development programs to adapt to new needs and gaps that are emerging in our socialization processes (e.g., onboarding)? How/are my junior hires included in educational work experiences that cannot be replicated by ChatGPT (e.g., client meetings, presentations)?

7. Managing a ChatGPT empowered workforce: Key takeaways

We have shown what ChatGPT affords workers, and we have also foreshadowed side effects where tactful managerial intervention is important. Potential interventions include:

- Repairing and retaining knowledge ties disrupted by ChatGPT.
- Facilitating employees and managers sharing their experiences with this technology.
- Offering upskilling opportunities or mentor programs in relation to ChatGPT.
- Creating awareness for employees that their individual learning—and as a result, organizational learning—is changing through ChatGPT and instruct them on how to collaborate appropriately with ChatGPT (e.g., verification of outputs becomes a significant task, work roles shift to being an editor or director).

8. Limitations and opportunities for future research

This article provides a snapshot of a rapidly developing technology: ChatGPT. Yet, we identified convergence in the use activities that were reported to us, which suggests that a degree of closure (Bijker et al., 2002) has occurred at least in the short term. While writing this article, OpenAI has launched new updates to ChatGPT (e.g., Heath, 2023). The updates potentially open up new affordances not included in this article. In the meantime, we offer a categorization of affordances at this point in time to reveal their implications and orient managers toward interventions.

Further, this study zoomed in on early adopters, but we recognize that employees who have not yet used ChatGPT may be more avoidant (Beaudry & Pinsonneault, 2005) than our insights reveal. For managers, it will be important in the future to oversee and manage this potential polarization in their workforce where some of the employees become very skilled with ChatGPT use in their work while others, ignoring it, miss out on gaining the skills needed to handle the technology.

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